MILITARY SPECIFICATION

COVERS, MARINE HATCH, HYDRAULICALLY OPERATED, QUICK ACTING, SELF-STOWING

1. SCOPE

1.1 Scope. - This specification covers self-stowing type, quick acting, hydraulically operated cargo hatch covers for use on the weather decks of auxiliary vessels.

1.2 Classification. - The hatch covers shall be of the following types, as specified (see 6.1):

Type I. - Raised type hatches. -
Type II. - Flush type hatches. -

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-621 - Boxes, Wood, Nailed and Lock-Corner.
NN-B-631 - Boxes, Wood, Wirebound (for Domestic Shipment).
PPP-B-601 - Boxes, Wood, Cleated-Plywood.

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JAN-P-106 - Packaging and Packing for Overseas Shipment-Boxes; Wood Nailed.
MIL-B-107 - Boxes, Wood, Wirebound (Overseas Type).
MIL-P-116 - Preservation, Methods of.
MIL-D-963 - Drawings, Production (for Electrical and Mechanical Equipment for Naval Shipboard Use).
MIL-F-3541 - Fittings, Lubrication (Hydraulic).
MIL-F-5504 - Filters, Hydraulic, Aircraft.
MIL-L-10547 - Liners, Case, Waterproof.
MIL-L-15017 - Lubricating Oil, Hudraulic.
MIL-I-15024 - Identification Plates, Information Plates and Marking Information for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-T-15071 - Technical Manuals for Mechanical and Electrical Equipment.
MIL-P-15130 - Paint, Outside, Haze-Gray, No. 27 (formula No. 5H) (for Naval Shipboard Use).
MIL-R-15137 - Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-C-15328 - Coating (Pretreatment) (formula No. 117 for Metals).
MIL-P-15424 - Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.
MIL-P-16789 - Preservation, Packaging, Packing and Marking of Pumps General, and Associated Repair Parts.
MIL-P-17545 - Primer, Paint, Shipboard, Alkyd-Red Lead Type, Formula No. 116.
MIL-P-17652 - Preparation for Delivery of Electronic, Electrical and Electro-Mechanical Parts.
MIL-C-17795 - Cleaning and Protecting Integral Piping for Oil Hydraulic Power Transmission Equipment.

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NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

MILITARY
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE
Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

AMERICAN INSTITUTE OF ELECTRICAL ENGINEERS
Bulletin No. 45 - Recommended Practice for Electrical Installations on Shipboard.

(Application for copies may be obtained from the American Institute of Electrical Engineers, 33 West 39th St., New York, N. Y.)

3. REQUIREMENTS

3.1 Cargo hatch covers shall be in accordance with the drawings listed in the contract or order and as specified herein.

3.2 General. - All cargo hatch covers shall be hydraulically operated, quick-acting, hinged, single or multi-pontoon section, self-stowing type. They shall be simple in operation and easily operable without the necessity of extensive preparation. The term "quick-acting" means that the hatch is capable of being opened from a completely secured condition, or reversely, completely secured from an open condition within a period of 6 minutes by one man without the use of any ship's cargo gear. These operations shall occur without placing undue stress on either the hatch cover or the coaming.

3.2.1 Degree of tightness. -

3.2.1.1 Watertight. - Watertight hatch covers shall be designed to be watertight under a hose test, with hose pressure of 50 p.s.i.g. The nozzle shall have a diameter of 1/2 inch and shall be not over 10 feet from the point under test.

3.2.1.2 Nontight. - Nontight hatch covers are not required to prevent or retard the passage of any fluid, and gaskets are not required.

3.2.2 All watertight hatch covers shall have their gaskets protected from damage due to loading operations in both the closed and open positions. No gasket drag or chafing is permissible. Hydraulic jacking devices shall be provided to prevent chafing of gaskets on all covers.

3.2.3 Flush lifting eyes or fittings shall be provided for each hatch cover to allow emergency operation of covers by means of ship's gear. All hatch covers shall be designed to minimize protuberances, gaps or pockets. All deck hinges or parts that have wearing surfaces requiring lubricating fittings shall have oil or grease groves machined in the bearing surfaces.

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3.2.4 In the construction of hatch covers all sharp and rough edges shall be ground smooth. The bottom edge shall be straight and true.

3.3 Operation. - Unless otherwise specified in the contract or order, in the design of the hatch covers, the contractor shall assume that the following conditions will prevail:

(a) Roll 10 degrees port to 10 degrees starboard in 6 seconds.
(b) Pitch 4 degrees up to 4 degrees down in 4 seconds.

3.3.1 The contractor shall supply preliminary drawings; criteria and proposal shall state the methods of operations, stowage location and space required, clear hatch opening, weights of all equipment and power requirement.

3.3.2 Calculations. - Calculations substantiating the selection of motor power, scantlings of hatch covers, stress analysis, pump capacity of system, operating pressure, cylinder size, and similar characteristics, shall be submitted to the bureau or agency concerned for approval. Scantlings of hatch covers shall be based on a design loading of 500 pounds per square foot and a deflection not greater than 1/360 of the span.

3.4 Hardware and fittings. -

3.4.1 Latching assemblies. - Latches shall be provided on the hatch cover in place of screws, bolts and dogs to secure the hatch covers in a minimum time without the use of special tools. Those sections which are attached to the coaming and pontoon by welding shall be free from any irregularities. The latches shall be of the quick-opening and quick-closing type. When the latch is engaged a "metal-to-metal" contact shall be accomplished between the hatch cover and sealing rail. These latches shall be of such design as not to disengage when the covers are loaded. Latch design shall allow one man to undog from a completely secured cover in not more than 3 minutes.

3.4.2 Hinges. - Hinges shall be furnished as an integral part of the hatch cover for attachment to the deck or hatch cover sections or both. These hinges shall be steel. They shall be secured with an SAE-1040, or equal, steel hinge pin, securely fastened. Each assembly shall be fitted with manganese bronze bushings, SAE-430, grade B. Lubrication fittings in accordance with Specification MIL-F-3541 shall be provided to insure proper lubrication to all working parts of the hinge.

3.4.3 Wheel assembly. - Multiple section hatches shall be fitted with wheels which shall be 1020 steel, or equal, and have a machined finish on the rolling surface. The wheels shall be designed so that the hatch cover may be opened in a rolling sea. Wheel assemblies shall be fitted with SAE-430B bronze bushings. The pins shall be of 1040 steel, or equal. Each assembly shall be fitted with lubrication fittings in accordance with Specification MIL-F-3541. Nuts on wheel assembly shall be corrosion resistant and lock wired. All wheel assemblies shall be readily accessible for lubrication, maintenance and inspection.

3.4.4 Gaskets, watertight. - The gaskets shall be installed in a gasket retainer section which shall be furnished as an integral part of each hatch cover section. The gasket shall be so arranged to permit ready removal by tools normally found aboard ship. The operation of all covers shall be designed so that no gasket drag, chafing or shear or both will occur. The gasket seat shall be protected from cargo damage and chafing. The gaskets shall be made from synthetic rubber, or a rubber-like material, having medium oil resistance. The gaskets shall be designed for absorbing deformation, encountered in the sealing action and for a long life, without taking a permanent set.

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3.5 Hydraulics. -

3.5.1 General. - The hydraulic system for the supply and control of fluid power shall be designed so as to conform to the following:

(a) All covers shall be operable individually with a maximum time limit for either opening or closing operations of 3 minutes per cover (without latching time).
(b) All hatch covers shall be able to be opened and closed simultaneously twice per hour under normal operating conditions.

3.5.1.1 The maximum working pressure shall be 3000 pounds per square inch (p.s.i.) and the hydraulic fluid shall be of a nature to provide normal operations in severe temperature ranges (minus 20° to plus 130°F.). The hydraulic system shall consist of three major items:

(a) Two hydraulic power units, one of which shall be located forward and the other aft, in a protected area on the main deck.
(b) Control stations which shall be located near and within full view of each hatch cover.
(c) The hatch cover actuating system.

3.5.1.2 The material for any hydraulic equipment shall not be cast iron unless specifically called for within this specification. Valve bodies and pump housings may be ductile iron or Meehanite or equal. Valves and controls shall be of the highest quality and designed for no internal leakage. The hydraulic system shall be thoroughly cleaned in accordance with Specification MIL-C-17795.

3.5.2 Power station. -

3.5.2.1 Operational requirements. - There shall be two duplicate power stations. One will supply power to the three hatch cover control stations in the forward half of the ship and the second power station will supply power to the two aft hatch cover control stations. Operation of all covers aboard ship through either power station shall be possible. The various connecting lines and valving arrangements shall be such that oil will flow from, and return to, the same power station when it is being used. Hydraulic valving will be incorporated in the hydraulic system to allow for the isolation of any part thereof. Each power station shall contain all equipment and piping necessary to reserve oil, pressurize oil and transmit it to the control stations. Means shall be provided to limit maximum pressure possible in the hydraulic system to 110 percent of the working pressure. The hydraulic system shall be designed to start and stop the pumping unit automatically as the system requires pressure. A sump reservoir or storage tank of adequate size shall be included with each power station. A means shall be provided within the power station and within the control stations to allow strained or filtered oil to be circulated through the hydraulic system while operating the hatch covers by means of ship's gear.

3.5.2.2 The power stations will be a combination of separate package units, completely piped for easy interconnection aboard ship.

3.5.2.3 Pumps. - The hydraulic pump shall be of the positive displacement, self-priming, self-lubricating type and of such a size as to perform the outlined functions.

3.5.2.4 Motors, electric. - Electric motor shall be furnished for each hydraulic power unit and shall be of the marine type, drip proof protected, 50°Centigrade (C.) ambient, 40° C. rise, 230 volts, direct current (d.c.), Class "A" insulation, horizontal, ball bearing, high starting torque, normal starting current. End shield and frame of motor shall be steel.

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3.5.2.5 Motor mount assembly. - All motors and pumps shall be furnished as operable units, mounted on a vase and connected to each other by means of a high strength steel flexible coupling.

3.5.2.6 Starters, electric. - The starters shall be across-the-line, magnetic type, drip proof protected, Marine in accordance with AIEE Bulletin No. 45, fused line disconnected, externally controlled.

3.5.2.7 Hydraulic oil filter. - An oil filter shall be furnished for each hydraulic power plant and installed aboard ship adjacent to the reservoir. The filter shall be the replaceable element micronic reservoir type in accordance with Specification MIL-F-5504.

3.5.2.8 Control panel. - A control panel unit shall be provided at each power station. This unit shall contain all required valves and gages.

3.5.3 Control boxes. -

3.5.3.1 Weatherdeck control box. - A separate control box shall be supplied for each hatch cover. These control boxes shall be located within the immediate vicinity of each hatch. The weatherdeck control box shall be of heavy sheet steel, all welded, weathertight construction. The instrument panel within the box and the box itself shall be of such design so as to allow servicing of all equipment within the box without removal of the equipment. All external hardware shall be brass or bronze. A fitting shall be provided to allow locking of the box. The equipment on an enclosed instrument panel, within the control box, shall control the supply of hydraulic power to the hatch cover sections. All valves shall be properly identified with brass identification plates in accordance with Specification MIL-I-15024. Operating and lubricating instructions of the hatch covers on lamicoid shall be attached to the inside cover of the control box.

3.5.3.2 Control valves. - The control valves which are located in the control boxes shall be of the four-way, closed center, panel mounted, sheer seal type, for hydraulic oil operation between 0 and 3000 p.s.i. The valve housings shall be of Meehanite, or equivalent, steel. They shall be designed for a minimum of leakage. These valves shall be able to withstand a proof pressure of 4500 p.s.i. against all ports except the return or reservoir port.

3.5.3.3 Pressure gauges. - All hydraulic pressure gauges shall be 0 to 5000 p.s.i., 4-1/2 inches dial, bourdon tube type, drawn steel or brass case construction. Each gauge shall have a gauge shut-off valve upstream in the system so as to allow taking the gauge out of the circuit and snubbing vibrations in the needle. All control boxes shall be fitted with pressure gauges.

3.5.4 Hydraulic system within the hatch cover sections. -

3.5.4.1 General requirements. - On all multiple section covers the hydraulic actuating means shall be completely contained within the hatch cover sections.

3.5.4.2 Actuation. - The hatch cover actuating system which shall be located inside the cover sections shall provide positive control of all sections of either half or single section of hatch covers at all points of travel. This hydraulic system also shall incorporate an automatic means of locking the hatch covers at all points of travel. The actuators shall be constructed and designed so as to provide adequate protection from the elements under all normal sea conditions. The actuator shall be pivot mounted in such a manner as to allow none but axial loads to be transmitted. Lubrication fittings shall be mounted at each end to lubricate the pivot mounting device. All actuators shall be statically tested at proof pressure. No visible leakage or porosity shall be allowed at this pressure and no undue stresses or deformation shall result to the actuator. Proof pressure shall be 4500 p.s.i. All hydraulic actuators shall also be tested under 3000 p.s.i. operation with an external load.

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3.5.4.3 Hydraulic tubing. - All hydraulic tubing shall be of steel and shall be capable of withstanding twice the maximum allowable working pressure. The tubing shall be seamless, cold drawn, and annealed AISI 1010, or equivalent. The tubing shall be smooth, clean and free from heavy scale, burns, tears, grooves, lamination, slivers, pits, and other injurious defects. The tubing shall conform to the following physical requirements:

Minimum tensile strength - 55,000 p.s.i.
Elongation in 2 inches - 30 percent minimum.

3.5.4.3.1 The finished tubing shall not crack or show any flaws when flattened between parallel plates to 3 times the wall thickness of the tube. Tubing shall withstand being expanded at one end over polished tapered mandral having an included angle of 60 degrees until the actual average inside diameter is increased 30 percent. All tubing, after installation, shall be subjected to a proof pressure of 4500 p.s.i. during which no leakage shall result.

3.5.4.3.2 All tubing shall be secured with a clamp that shall be cadmium plated and cushioned with a silicate compound around the tubing grip section. The clamps shall be spaced no more than 3 feet apart and at all bends.

3.5.4.4 Hydraulic fittings. - The fittings which shall be used to connect the sections of steel tubing shall be of a compression type satisfactory to the bureau or agency concerned. They shall be satisfactory to the Coast Guard for the working pressures involved. They shall be of the flareless type, capable of withstanding working pressures of 3000 p.s.i. and shall be capable of withstanding shock loads, vibration with positive sealing and proof pressure of 4500 p.s.i. All fittings shall be manufactured of C-1117 steel, Barstock or C-1137 steel forgings, or equivalent. The term "fittings" as used in this specification shall refer to reducers, male connectors, female connectors, unions, bulkhead unions, bulkhead tees, 45 degree elbows, 90 degree elbows, tees, crosses, caps, adaptors, plugs and terminals.

3.5.4.5 Flexible hose. - The interconnection of the steel tubings between hatch cover sections and hatch cover to hydraulic actuating mechanism shall be accomplished with flexible hose which shall be designed to eliminate, insofar as possible, the fatigue factor. Each end of the interconnecting hose shall be terminated at a fixed vibration-proof fitting. Entrance of hydraulic pressure from the power station into the hatch cover sections shall be accomplished by a length of flexible hose from the deck to hatch cover. The hose shall follow the axis of articulation of the hinges connecting the deck to the hatch cover sections. The flexible hose shall be constructed around a seamless synthetic rubber innertube surrounded by two high tensile wire braids bonded with plies of synthetic rubber. The outside cover shall be constructed from tough oil resistant rubber which has been impregnated with cotton braids. The hose shall be able to withstand a temperature between minus 40° to plus 200° Fahrenheit (F), and be able to withstand maximum working pressure of 4000 p.s.i., a proof pressure of 4500 p.s.i. and shall have a minimum burst pressure of 15,000 p.s.i.

3.5.4.6 Automatic emergency control. - There shall be provided a means for controlling the rate of descent of the hatch covers in the event that one of the hydraulic lines becomes severed.

3.5.5. Hydraulic oil. - The hydraulic oil required in the hydraulic system shall be Military symbol MS 2075H, in accordance with Specification MIL-L-15017.

3.6 The covers and hydraulic equipment for their operation shall be furnished by the contractor, complete and ready for installation on shipboard including the following:

(a) Hydraulic tubing attached to cover, pumping units and control boxes.
(b) Hydraulic pump and electric motor assembled and mounted as a unit.
(c) Electrical equipment such as starters, controllers, pressure switches, and limit switches.
(d) Hydraulic fluid.
(e) Paint as specified in 3.9.

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(f) Lifting eyes.
(g) Latching devices, complete.
(h) Control boxes assembled and complete.
(i) All hydraulic equipment required at power station.
(j) Control panel for power station equipment.
(k) Repair parts.
(ℓ) Tools.

3.7 Type I. - Type I hatches shall close upon a coaming, the top of which is above the top of the deck, on which it is installed.

3.8 Type II. - Type II hatches shall be so fitted that the hatch covers when closed are flush with the top surface of the deck on which it is installed.

3.9 Painting. - Cargo hatch covers shall be painted with one coat of wash primer, formula No. 117, in accordance with Specification MIL-C-15328, two coats of red lead primer, formula No. 116, in accordance with Specification MIL-P-17545, and two coats of haze gray, formula No. 5H, in accordance with Specification MIL-P-15130. Prior to painting, mill scale on both sides of the hatch covers shall be removed by sandblasting or pickling or both.

3.10 Drawings. - The contractor shall prepare assembly and working drawings, including diagrams, schematic of hydraulic circuits, arrangements, details and sketches, bills of materials defining material and equipment. Drawings shall be submitted to the bureau or agency concerned for approval. Drawings shall be pen-and-ink on cloth, or photographic tracings on cloth. They shall be in accordance with Specification MIL-D-963 containing sufficient detail to permit fabrication of any part listed.

3.11 Technical manuals. - Technical manuals shall conform to type C of Specification MIL-T-15071. The manuals shall include reduced size prints of the approved outline and sectional drawings of all equipment.

3.12 Repair parts. - Complete sets of shipboard repair parts, as recommended by the hatch cover contractor for 1 year's service, shall be furnished by the contractor in accordance with Specification MIL-R-15137.

3.12.1 Repair parts shall include at least the following:

2 - Springs of each size and kind.
2 - Grease fittings of each size and kind.
1 - Complete set of special nuts, bolts, screws, collars, dogs, shim materials, and similar items.
1 - Gasket of each size and kind for one complete hatch of maximum size.

3.13 Special tools. - Special tools shall be furnished by the contractor.

1 - Set of wrenches as recommended by the contractor.
1 - Grease gun per hatch cover.

Special tools are defined as those tools not listed in the Catalog of Naval Material, General Stores Section. (Copies of this catalog may be consulted in the office of the Government inspector.)

3.14 The contractor shall furnish the services of a competent engineer during testing and installation of covers and pumping units as required.

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3.15 Workmanship. - The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Tests. - At the time of fabrication and before installation, all watertight hatch covers shall be subjected to a watertightness test. All equipment for the power stations shall be shop tested by the contractor prior to installation aboard ship. After installation aboard ship, all watertight hatch covers shall be hose tested. Prior to acceptance of watertight hatch covers after manufacture, a complete hydraulic working test will be conducted on each watertight hatch cover at the manufacturer's plant to assure proper alignment and operability of units. The pressure-limiting device (see 3.6.2.1) shall be tested at 110 percent of working pressure at least 15 times. This pre-test shall simulate shipboard operation and shall result in a minimum of shipyard adjustment during installation. The contractor shall submit for approval estimated maximum deflection calculations. Each cover shall be operated successfully through ten cycles of opening and closing without malfunction or leaks.

4.2 Inspection procedures. - For naval purchases, the general inspection procedures shall be in accordance with General Specification for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. -

5.1.1 For domestic shipment-immediate use. - The hatch covers and operating mechanism shall be preserved and packaged in accordance with commercial practice to afford protection to the equipment in transit, temporary storage incident to installation and during installation.

5.1.2 For domestic-shipment and storage or overseas shipment. -

5.1.2.1 Equipment Hatch covers and operating mechanism shall be preserved and packaged in accordance with the methods and materials requirements of Specification MIL-P-116 as detailed in the contract or order.

5.1.2.2 Repair parts and tools. - Repair parts and tools shall be preserved and packaged in accordance with Specification MIL-P- 17652 MIL-P-16789, MIL-P-15424 and in accordance with Specification MIL-P-116 for items not specifically covered therein. Parts and tools, generally, shall be packaged individually except when used in sets or quantities greater than one.

5.2 Packing. -

5.2.1 For domestic shipment-immediate use. - The hatch covers and operating mechanism shall be packed to insure carrier acceptance and safe delivery to destination at the lowest applicable rate. Containers shall comply with the Consolidated Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.2.2 For domestic shipment and storage or overseas shipment. -

5.2.2.1 Equipment. - The hatch covers and operating mechanism shall be packed for shipment as specified in the contract or order.

5.2.2.2 Repair parts and tools. - Repair parts and tools shall be packed in cleated plywood, nailed wood or wirebound boxes conforming to Specification PPP-b.601 (domestic types), NN-B-621 and NN-B-631 for domestic shipment and storage and conforming to Specification PPP-B-601 (overseas type), JAN-P-106 and MIL-B-107 respectively for overseas shipment. For overseas shipment, the boxes shall be lined with sealed caseliners conforming to Specification MIL-L-10547. The grass weight of wood or wood cleated boxes shall not exceed approximately 200 pounds.

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5.3 Technical manuals and drawings. - Technical manuals shall be packaged and packed in accordance with Specification MIL-T-15071. Drawings shall be prepared for shipment in accordance with Specification MIL-D-963.

5.4 Marking. - In addition to any special marking required in the contract or order, marking of the packages and shipping containers shall be in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Whether for domestic shipment-immediate use; domestic shipment and storage or overseas shipment (see 5.1 and 5.2).

Patent notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United Stated Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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